BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(MID SEMESTER EXAMINATION)

CLASS: BTECH
BRANCH: CSE

SEMESTER : VI
SESSION : SP/2026

SUBJECT: IT349 CRYPTOGRAPHY AND NETWORK SECURITY

TIME: 02 Hours

FULL MARKS: 25

INSTRUCTIONS:
1. The question paper contains 5 questions each of 5 marks and total 25 marks.
2. Attempt all questions.
3. The missing data, if any, may be assumed suitably.
4. Tables/Data handbook/Graph paper etc., if applicable, will be supplied to the candidates

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			CO	BL
Q.1(a)	Differentiate between Cryptography and Steganography with the help of a suitable example for each.	[2]	1	2
Q.1(b)	Illustrate and explain the Model for Network Security by constructing a suitable block diagram.	[3]	2	3
Q.2(a)	Distinguish between active and passive attacks with examples for each type of attack.	[2]	1	2
Q.2(b)	Explain the various security mechanisms in network security. How are these mechanisms related to security services?	[3]	1	4
Q.3(a)	Distinguish between monoalphabetic and polyalphabetic ciphers.	[2]	2	2
Q.3(b)	Encrypt the message "this is an exercise' using the Affine cipher with key (15,20). Ignore the spaces between the words while encrypting. Also decrypt the message to get the original plaintext.	[3]	2	3
Q.4(a)	Apply Euclidean Algorithm to determine gcd of(341,527) and express it as linear combination of 341 and 527.	[2]	1	3
Q.4(b)	Using Euler's Theorem, find the last two digits of 7^{45}. Show all necessary steps.	[3]	1	3
Q.5(a)	Define the congruence with its basic properties in modular arithmetic.	[2]	1	1
Q.5(b)	Using Fermat's Little Theorem, evaluate: $9^{793} \pmod{73}$ Show all necessary steps.	[3]	1	3

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